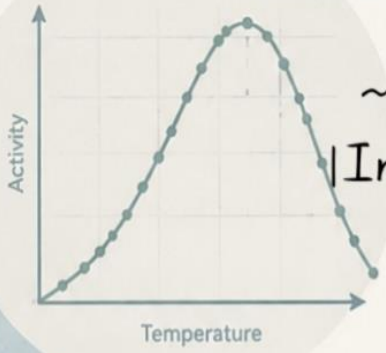
INVESTIGATION OF THE EFFECTS OF TEMPERATURE AND PH ON CATALASE ACTIVITY IN POTATO TISSUE

Independent Student Research · Biochemistry

⭐ Research Question: How do temperature and pH affect the activity of catalase in potato tissue?



~ By Avani Kaushik
|Independent Student Researcher|

ABSTRACT

This study investigated the effect of temperature and pH on the activity of the enzyme catalase present in potato tissue. Catalase catalyzes the decomposition of hydrogen peroxide into water and oxygen. Potato cubes were exposed to hydrogen peroxide under different temperature and pH conditions, and enzyme activity was assessed by observing oxygen bubble production. The results showed that catalase activity varied with environmental conditions. Maximum activity was observed at moderate temperatures and near-neutral pH, while both low and high temperatures, as well as acidic and alkaline conditions, reduced enzyme activity. These findings demonstrate that enzymes function most efficiently under optimum conditions and that changes in temperature and pH can significantly affect enzymatic reactions.

INTRODUCTION

The term ENZYME was proposed by **Wilhelm Kühne** in **1878.** Enzymes are soluble substances produced in living cells and are always of the nature of protein. They act as **biocatalysts**. They alter the rate of reaction without participating in the reaction themselves.



|Wilhelm Kühne|

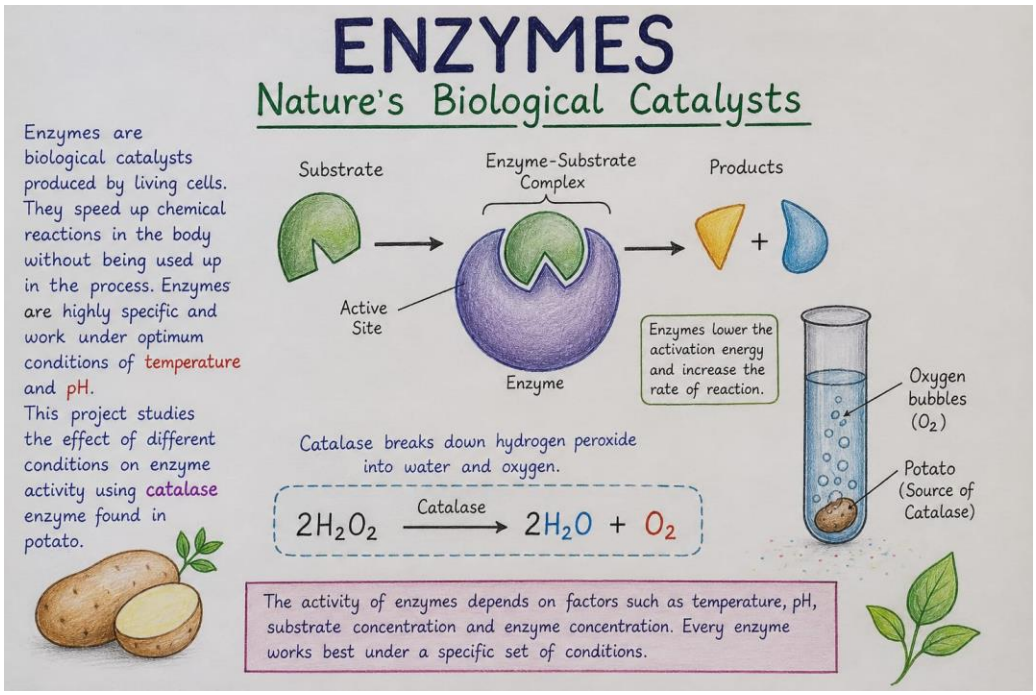
Enzymes are biological catalysts that speed up biochemical reactions occurring in living organisms without being consumed in the reaction. They are proteinaceous in nature and are highly specific in their action. Enzymes play an important role in digestion, respiration, photosynthesis, and other metabolic processes essential for life.

The activity of enzymes is influenced by several factors such as temperature, pH, enzyme

concentration, substrate concentration, and inhibitors. Every enzyme functions most efficiently under a specific set of conditions known as optimum conditions. Any change in these conditions affects the rate of enzymatic reactions.

This project aims to study the effect of temperature and pH on enzyme activity using catalase enzyme present in potato tissue.

LITERATURE REVIEW



- GENERAL PROPERTIES:

1) They are always produced by the activity of the living protoplasm.
2) They are effective in small quantities.
3) They are sensitive to heat and most of the enzymes are rendered useless at high temperature.
4) They are generally very specific in their action that is 1 enzyme can bring about only

one type of chemical change that too involving only one type of substrate.

5) Most but not all types of enzyme -catalysed reaction are reversible.

6)They require a specific pH for their optimum activity.

- NOMENCLATURE AND CLASSIFICATION OF ENZYMES:

  The most recent and rational system of enzyme classification as well as nomenclature is that proposed by the **International Union of Biochemistry (IUB).** It was initiated in 1972 and is periodically updated to incorporate discovered enzymes and new knowledge concerning catalytic mechanisms. This system is based on reaction types and their mechanisms.

  According to this system, all enzymes are grouped into six major classes and each class is divided into subclasses. All the known enzymes bear a characteristic code number

(that is EC), example, hexokinase is coded as E.C.,2.7.1.1. the major classes of enzymes are as follow:

1) Oxidoreductases- they participate in oxidation-reduction reactions.
   example -dehydrogenases, oxidases catalase, peroxidase.
2) Transferases- they catalyse the transfer of functional group (aldehyde, ketone, phosphate, sulphur ,etc) from one substrate to another.
   Example- hexokinase, amino transferases ETC.
3) Hydrolases-they catalyse hydrolysis of bonds by addition of water. All the digestive enzymes fall into this category.
   Example- amylase, trypsin, lipase etc.
4) Lyases- they catalyse removal of groups from the substrates by mechanism other than hydrolysis forming double bonds in the process.
   Example- aldolase, fumarase, carboxylase etc.

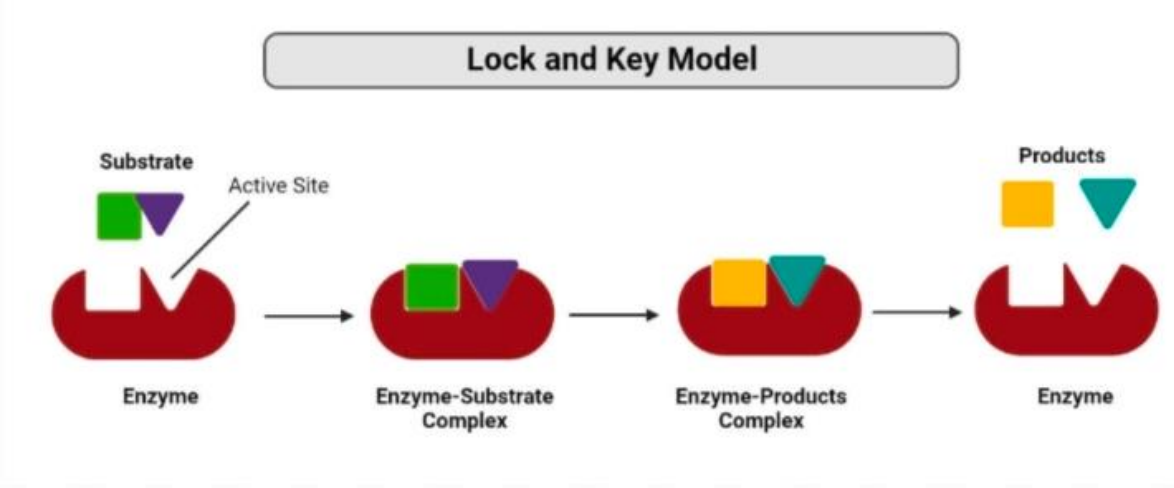
5) Isomerases- these catalyse intramolecular rearrangement resulting into formation of isomers.
   Example- phosphohexoisomerase, epimerase, alanine, racemase etc.
6) Ligases or synthetases- they catalyse synthesis by condensation of two groups requiring ATP or a similar triphosphate .
   Example- RNA synthetase, DNA ligase, glutamine synthetase etc.

- MECHANISM OF ENZYME ACTION

During the course of reaction an enzyme does not take a direct part in it. It merely provides a platform or template where reaction molecules come into contact much faster than chance collisions at that temperature. Their principal effect is to lower the activation energy of the reaction, e.g., decomposition of H_2O_2 without a catalyst requires 18,000 calories per mole which comes down to only 2,000 calories in presence of catalase.

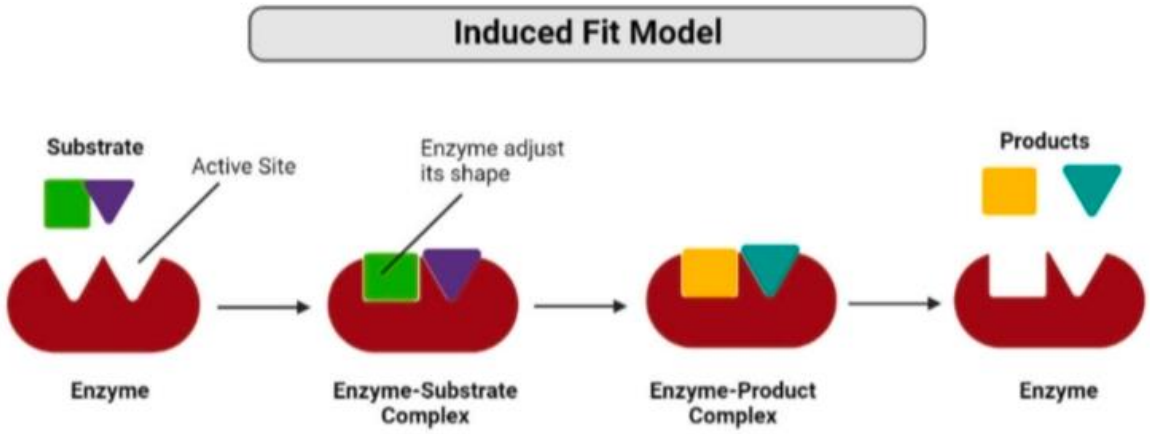
There are two different models to explain the mechanism of enzyme action:

1. Lock and key model



It was proposed by **Emil Fischer (1894).** According to this model, just as only a specially shaped key fits into a particular lock, similarly a specific catalytic site will accept only a substrate having a complementary shape. This model accounts well for the theory of specificity of enzymes. According to this model, active site is assumed to be pre-shaped to fit the substrate.

2. Induced fit model



This model was proposed by **Koshland (1960).** According to him, the shape of catalytic site of an enzyme is not pre-determined; rather it is the binding of substrate with the enzyme that induces a conformational change in the enzyme that brings the required amino acids in the correct spatial orientation for proper catalytic property. Thus, according to this model, shape of catalytic site is flexible. This model is supported by X-ray crystallographic studies which indicate a change in the shape of active site after binding with appropriate substrate.

THEORY

→Enzymes

Enzymes are biological catalysts produced by living cells. They increase the rate of chemical reactions by lowering the activation energy required for the reaction. Enzymes are highly specific and generally act on a particular substrate.

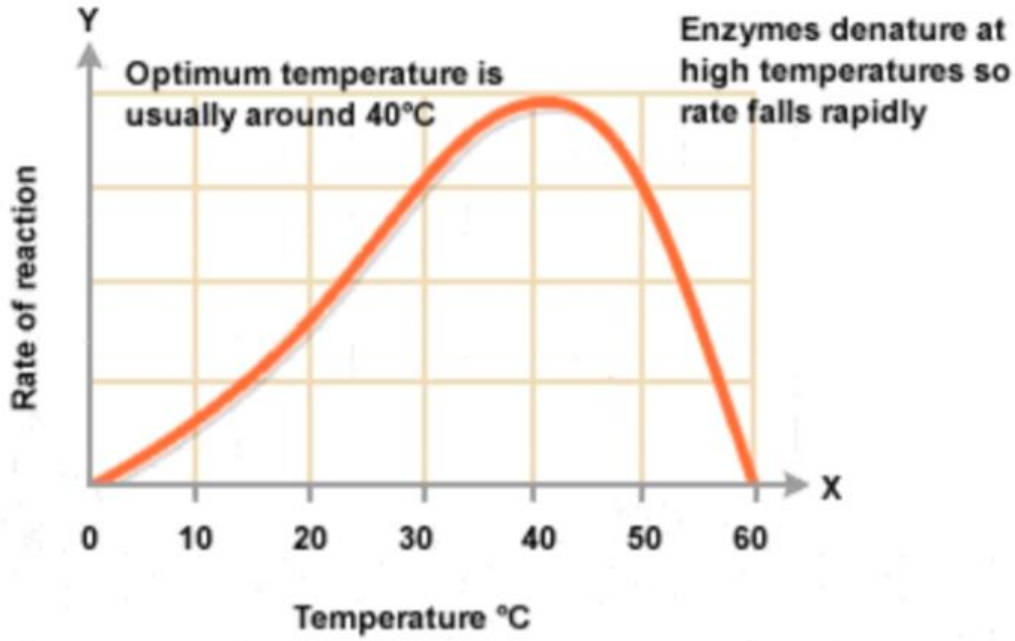
Catalase is an enzyme commonly found in potato cells. It decomposes hydrogen peroxide into water and oxygen.

Reaction:

$2H_2O_2 \rightarrow 2H_2O + O_2$

The oxygen produced appears as bubbles and serves as an indicator of enzyme activity.

→Effect of Temperature on Enzyme Activity



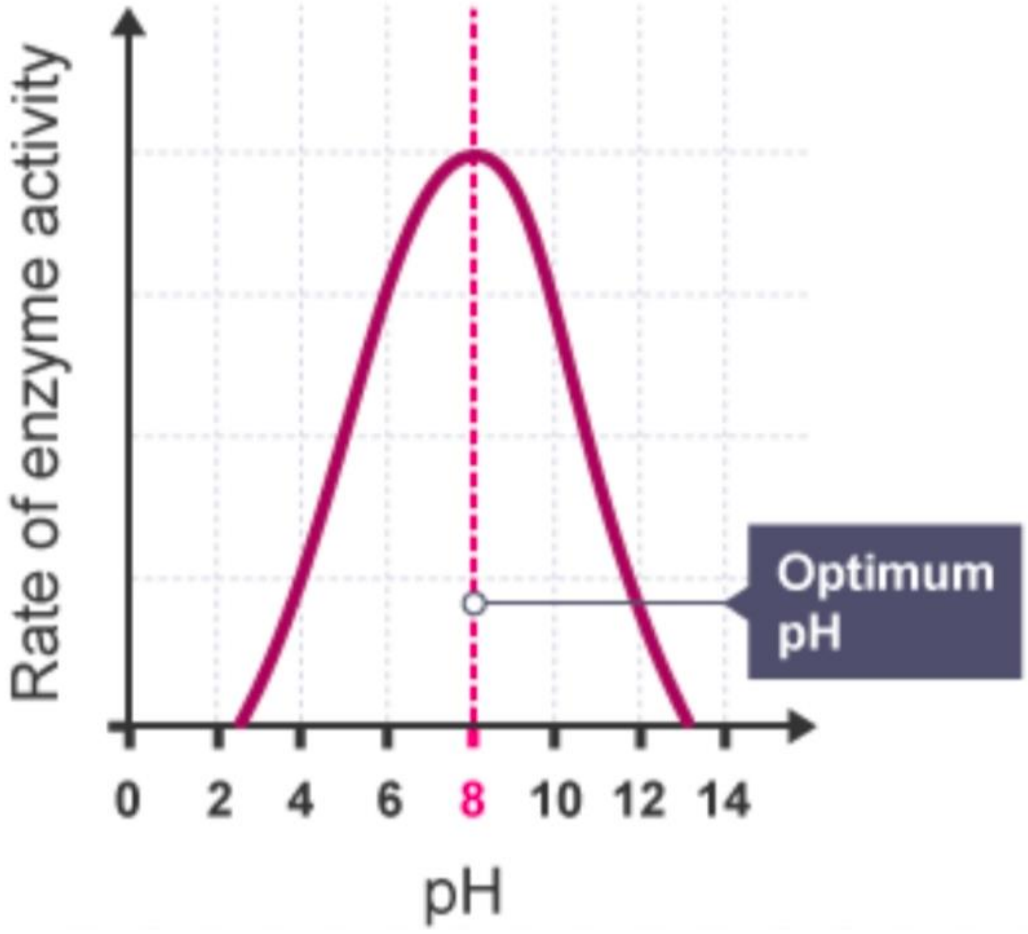
Temperature greatly influences enzyme activity. As temperature increases, the kinetic energy of molecules increases, leading to more frequent collisions between enzyme and substrate molecules. Therefore, the rate of reaction increases.

Most enzymes show maximum activity at an optimum temperature between 25°C and 45°C. Beyond this temperature, the enzyme structure becomes altered or denatured, causing a decrease in activity.

HYPOTHESIS:

At very low temperatures, enzyme activity is slow because molecular movement is reduced.

→**Effect of pH on Enzyme Activity**



Each enzyme functions best at a specific pH called the optimum pH. Changes in pH affect

the ionic bonds and three-dimensional structure of the enzyme.

→Examples:
- Pepsin works best at pH 1.5–2.5
- Salivary amylase works best near pH 7
- Trypsin works best between pH 8 and 11

HYPOTHESIS:
When the pH deviates from the optimum value, enzyme activity decreases.

EXPERIMENTS

OBJECTIVES:

1. To study the effect of temperature on enzyme activity.

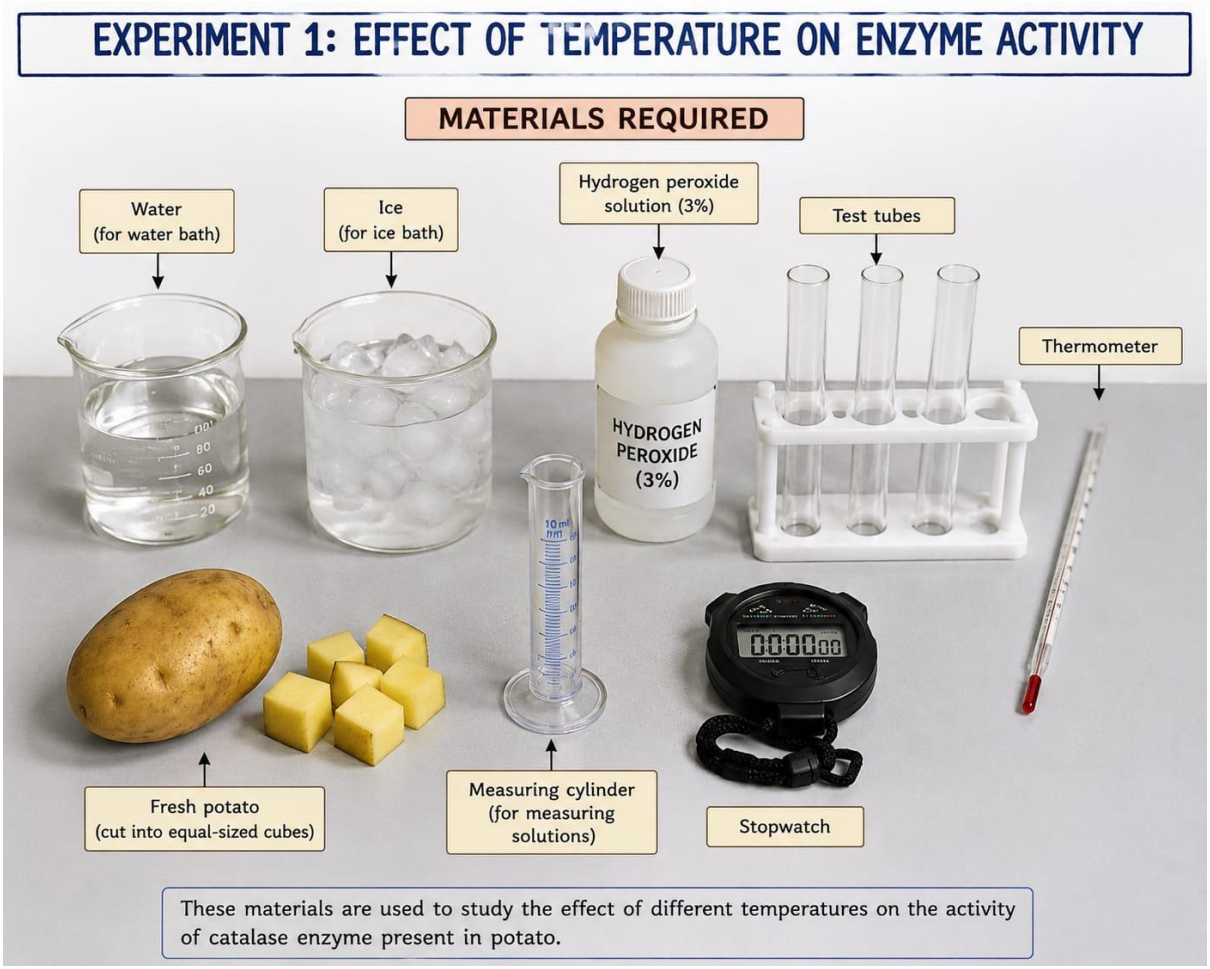
2. To study the effect of pH on enzyme activity.

3. To determine the optimum conditions required for enzyme activity.

4. To understand how environmental factors influence enzyme-catalyzed reactions.

EXPERIMENT 1: EFFECT OF TEMPERATURE ON ENZYME ACTIVITY

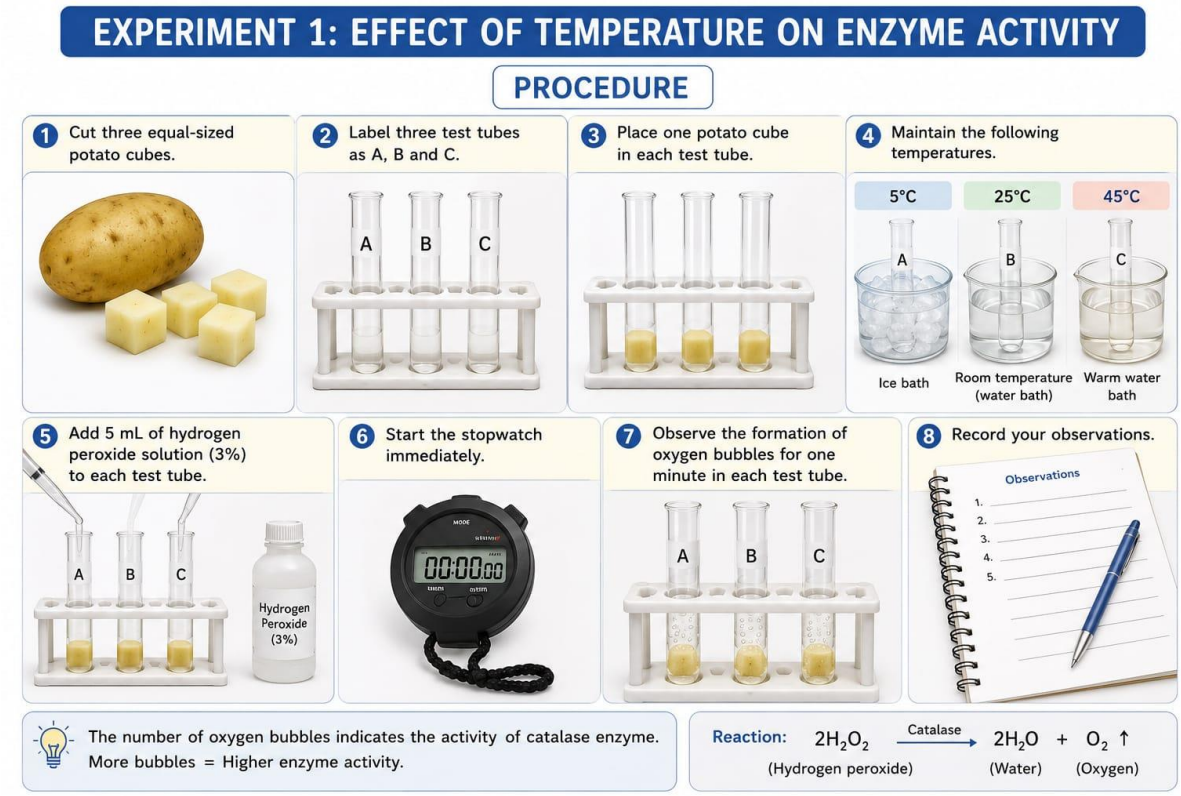
- Materials Required:



- Fresh potato
- Hydrogen peroxide solution (3%)
- Three test tubes
- Water bath

- Ice bath
- Thermometer
- Stopwatch

- Procedure:

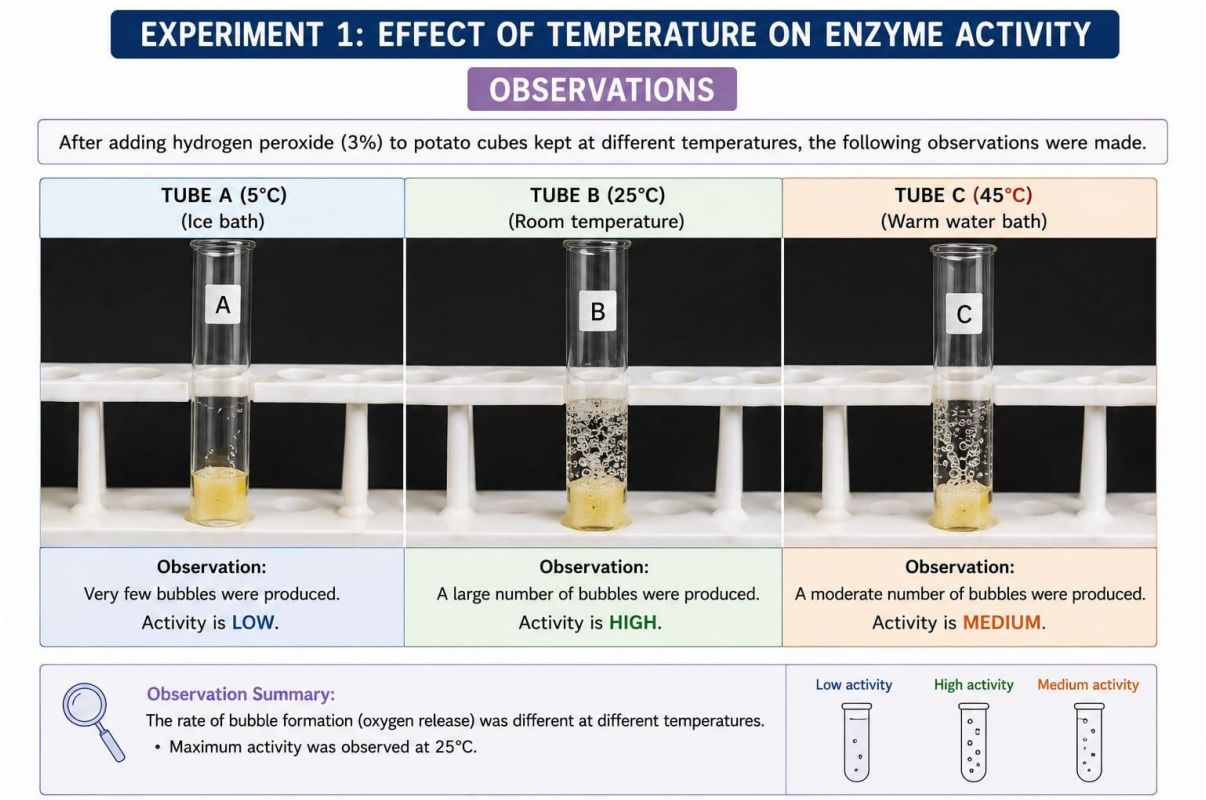


1. Cut three equal-sized potato cubes.
2. Label three test tubes as A, B, and C.
3. Place one potato cube in each test tube.
4. Maintain the following temperatures:
   - Tube A: 5°C

- Tube B: 25°C
- Tube C: 45°C

5. Add 5 mL hydrogen peroxide to each test tube.
6. Observe the formation of oxygen bubbles for one minute.
7. Record the observations.

- Observations:



The number of oxygen bubbles produced varied with temperature.

- Observation Table:

TUBE	TEMPERATURE	OBSERVATION	RELATIVE ACTIVITY
A	5°C	FEW BUBBLES	LOW
B	25°C	LARGE NUMBER BUBBLES	HIGH
C	45°C	MODERATE BUBBLES	MEDIUM

- Result:

Maximum enzyme activity was observed at 25°C.

- Conclusion:

Catalase activity increases with temperature up to an optimum level. At low temperatures, activity is slow, while higher temperatures reduce activity due to enzyme denaturation.

- Precautions:

1. Use equal-sized potato cubes.

2. Maintain accurate temperatures.

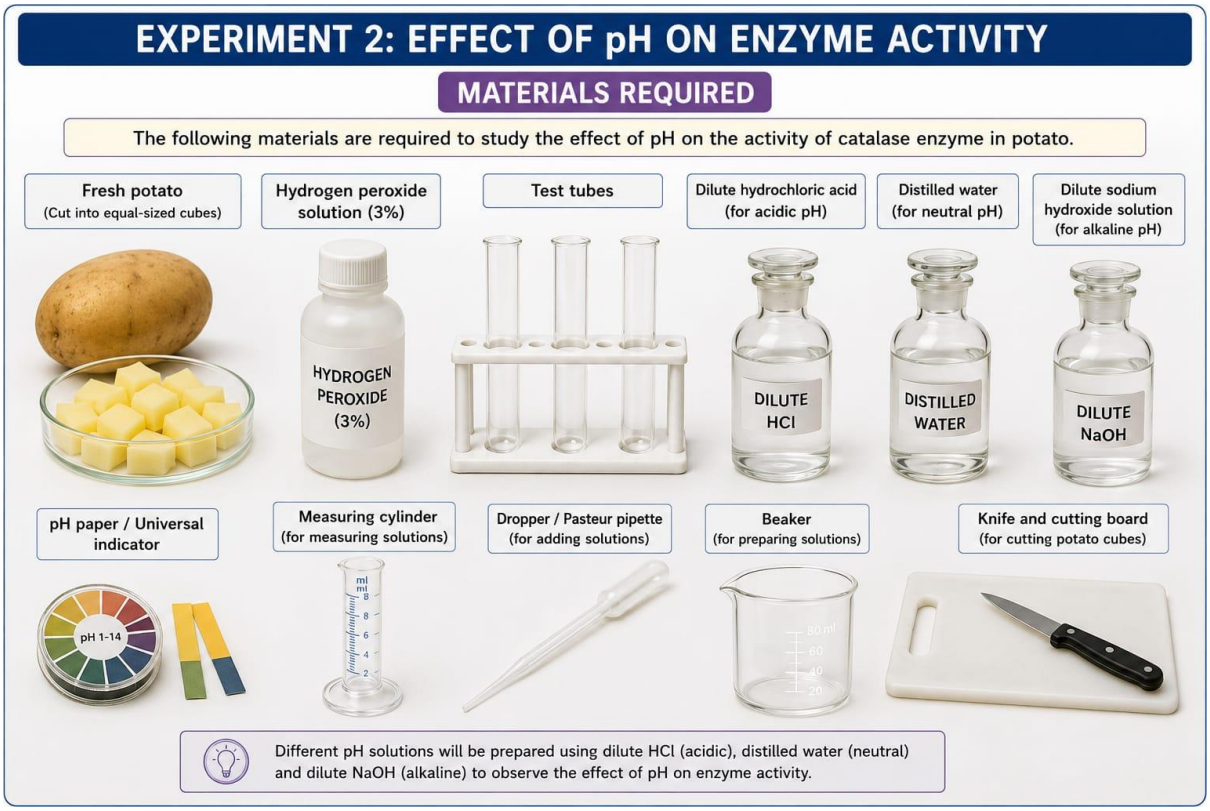
3. Use equal volumes of hydrogen peroxide.

4. Observe bubble formation immediately after adding hydrogen peroxide.

5. Use clean apparatus.

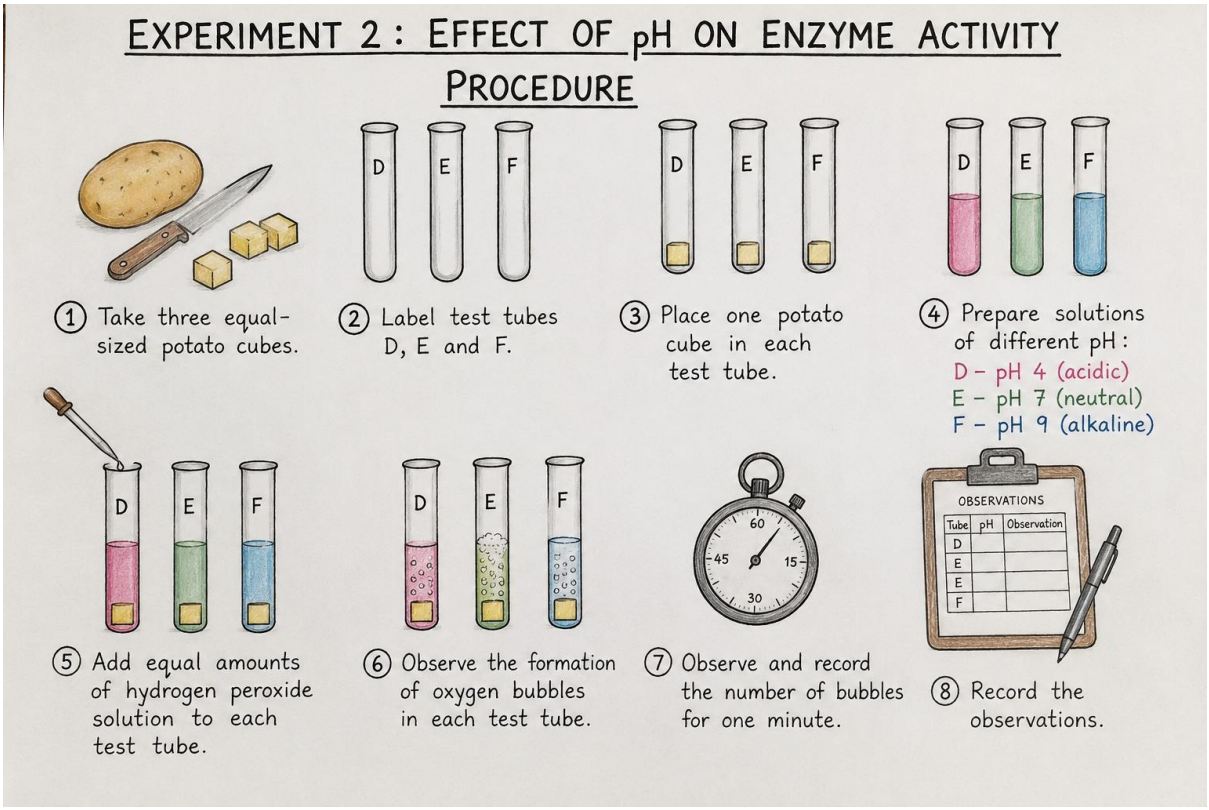
EXPERIMENT 2: EFFECT OF pH ON ENZYME ACTIVITY

- Materials Required:



- Fresh potato cubes
- Hydrogen peroxide solution (3%)
- Test tubes
- Dilute hydrochloric acid
- Distilled water
- Dilute sodium hydroxide
- pH paper

- Procedure:

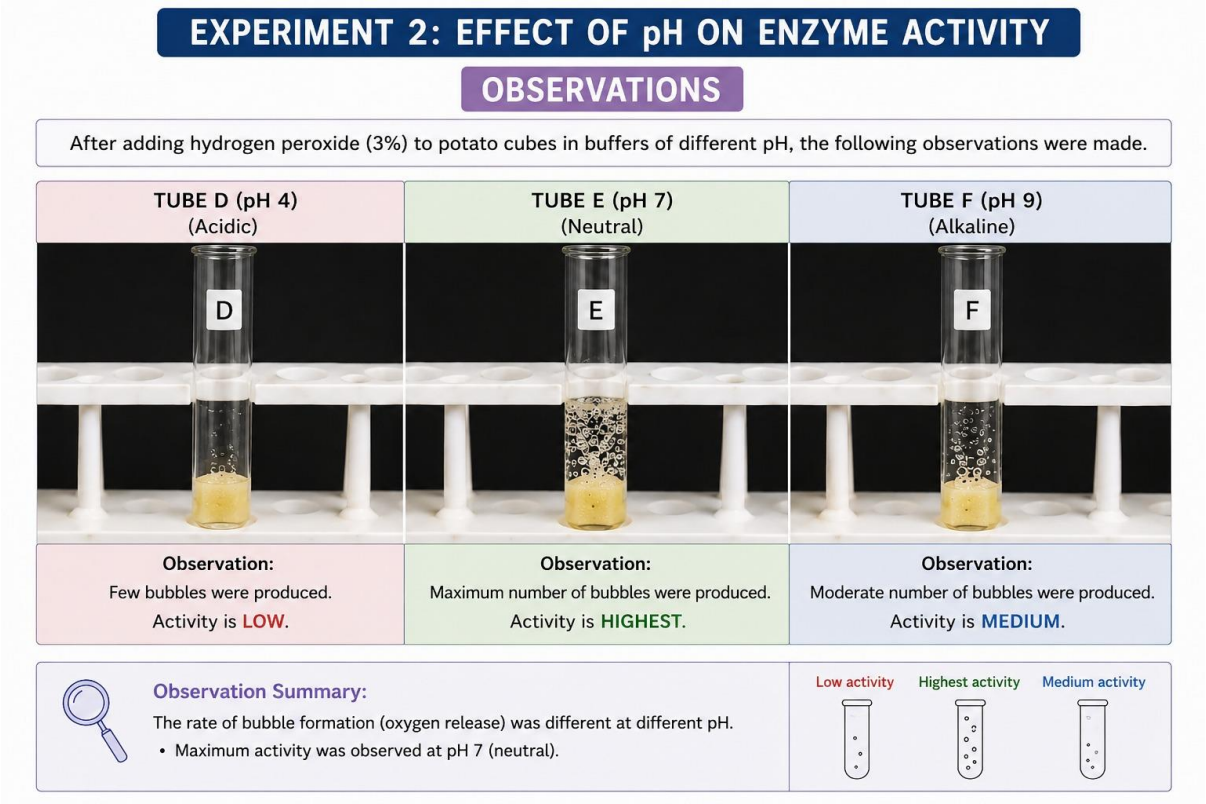


1. Take three equal-sized potato cubes.
2. Label test tubes D, E, and F.
3. Prepare solutions of different pH:
   - Tube D: pH 4
   - Tube E: pH 7
   - Tube F: pH 9
4. Place one potato cube in each test tube.
5. Add equal amounts of hydrogen peroxide.

6. Observe the formation of oxygen bubbles.

7. Record the observations.

- Observations:



The amount of oxygen bubbles produced varied with pH.

- Observation Table:

TUBE	pH	OBSERVATION	RELATIVE ACTIVITY
D	4	FEW BUBBLES	LOW
E	7	MAXIMUM BUBBLES	HIGHEST
F	9	MODERATE BUBBLES	MEDIUM

- Result:

Maximum enzyme activity was observed at pH 7.

- Conclusion:

Catalase functions most efficiently at neutral pH. Both acidic and alkaline conditions reduce enzyme activity.

- Precautions:

1. Maintain accurate pH values.
2. Use equal-sized potato cubes.
3. Use equal quantities of hydrogen peroxide.
4. Handle acids and alkalis carefully.
5. Use clean glassware.

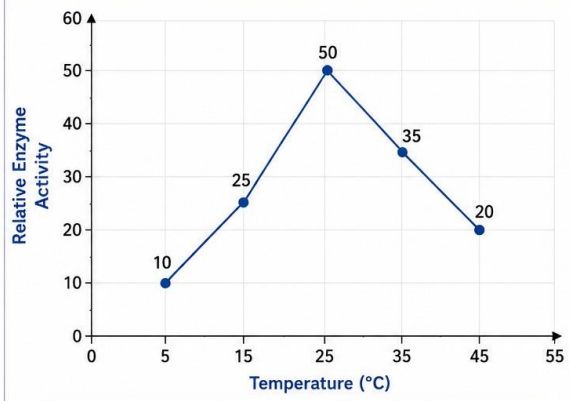
OVERALL RESULTS:

1. Enzyme activity is affected by temperature.
2. Enzyme activity is affected by pH.
3. Maximum catalase activity occurs under optimum conditions.
4. Extremely high temperatures and unsuitable pH values reduce enzyme efficiency.

COMPARATIVE GRAPH OF EXPERIMENT 1 & EXPERIMENT 2

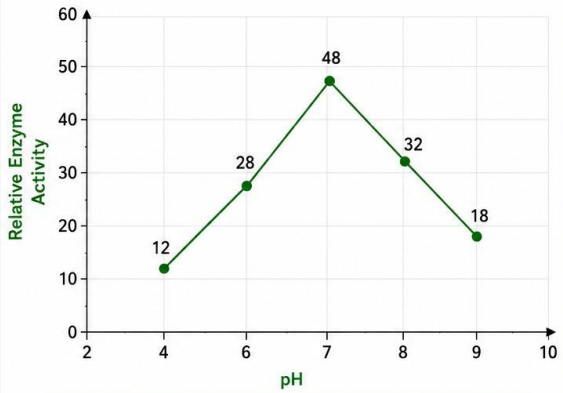
The following graph compares the effect of Temperature and pH on the activity of catalase enzyme (measured by the relative number of oxygen bubbles produced per minute).

EXPERIMENT 1: EFFECT OF TEMPERATURE



Maximum enzyme activity was observed at 25°C.
Activity increases with temperature up to the optimum point and then decreases.

EXPERIMENT 2: EFFECT OF pH



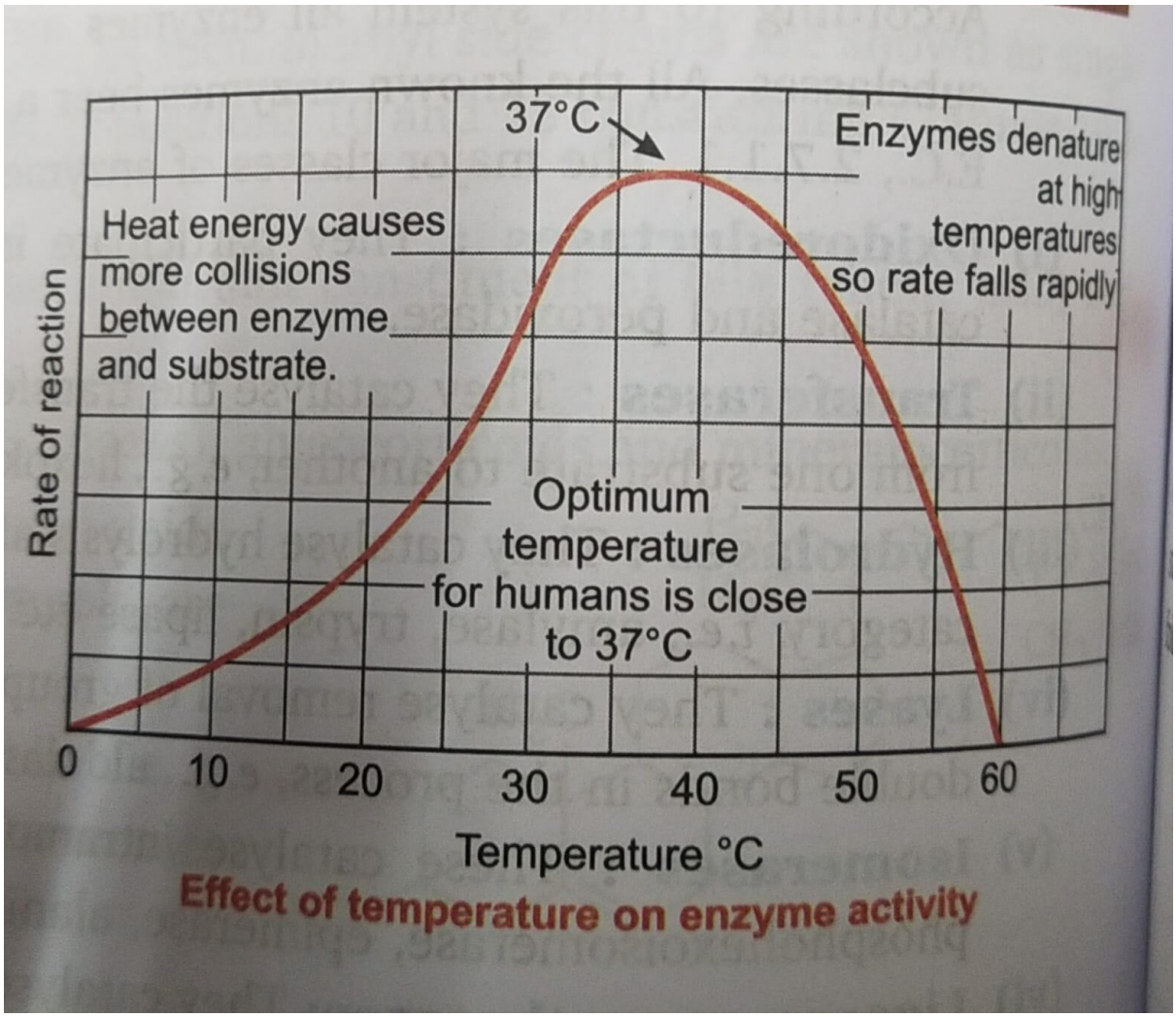
Maximum enzyme activity was observed at pH 7 (neutral).
Activity increases with pH up to the optimum point and then decreases.

COMPARISON:

- Both temperature and pH affect the activity of catalase enzyme.
- Enzyme shows maximum activity at an optimum temperature (25°C) and optimum pH (7).
- Beyond the optimum point, activity decreases due to denaturation or change in enzyme structure.

GRAPHS

Graph 1: Temperature vs Enzyme Activity



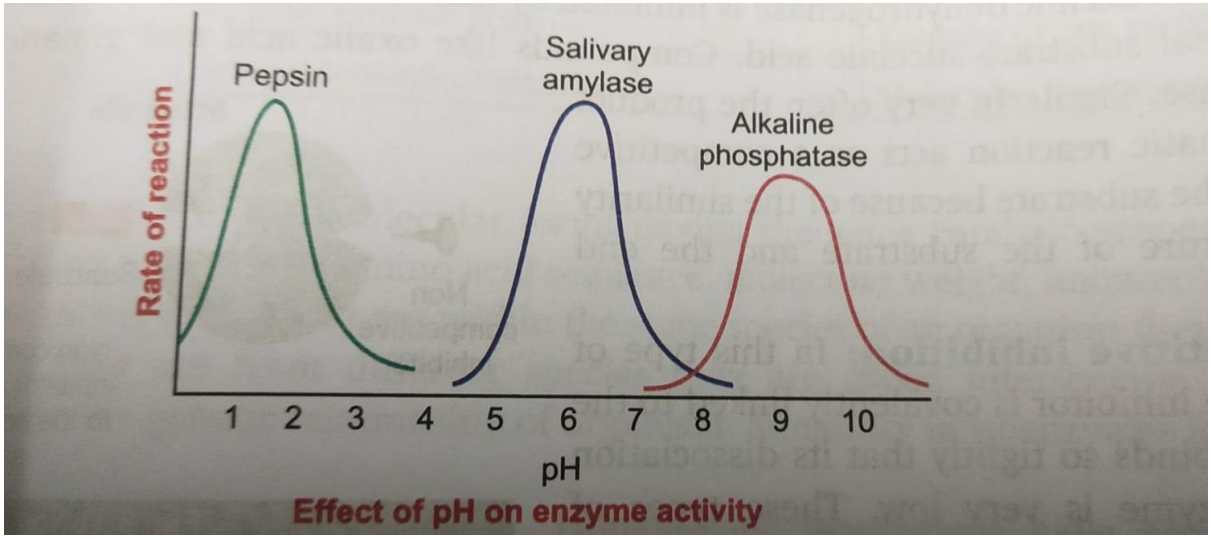
Plot:

- X-axis: Temperature (°C)

- Y-axis: Relative Enzyme Activity

The graph should rise gradually to the optimum temperature and then decline sharply.

Graph 2: pH vs Enzyme Activity



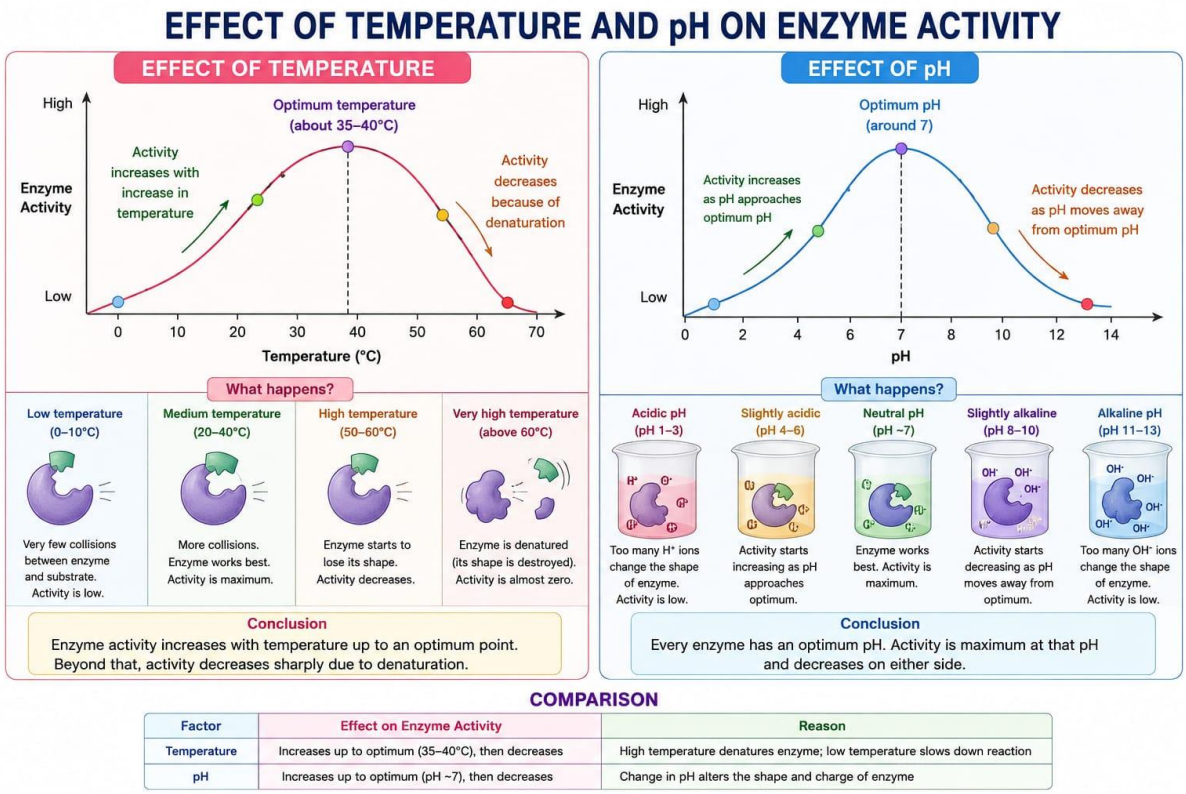
Plot:

- X-axis: pH

- Y-axis: Relative Enzyme Activity

The graph should be bell-shaped with maximum activity around pH 7.

COMPARITIVE STUDY:

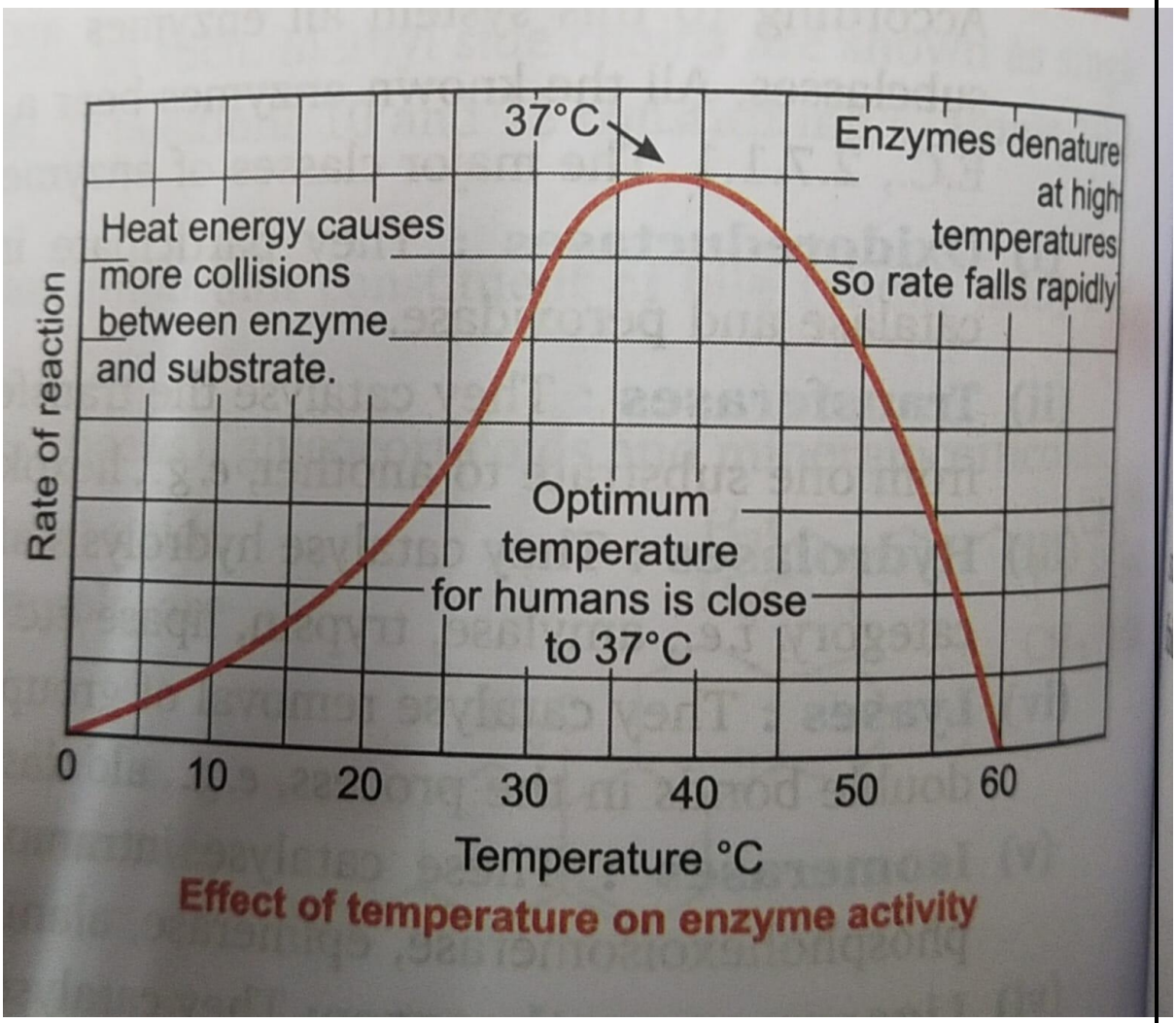


The graphs showing the effect of pH and temperature on enzyme activity are similar in that both are bell-shaped curves with a distinct optimum point where enzyme activity is maximum. On either side of the optimum pH or optimum temperature, enzyme activity decreases. However, the temperature graph usually shows a gradual increase in activity up to the optimum temperature followed by a sharp decline due to enzyme denaturation. In contrast, the pH graph is

generally more symmetrical, as activity decreases on both sides of the optimum pH because changes in pH alter the ionization of amino acids at the active site, affecting enzyme-substrate binding. Thus, both factors influence enzyme activity, but high temperatures can permanently denature enzymes, whereas pH changes mainly affect the active site's structure and charge.

FACTORS AFFECTING THE ENZYME ACTIVITY:

1. TEMPERATURE-

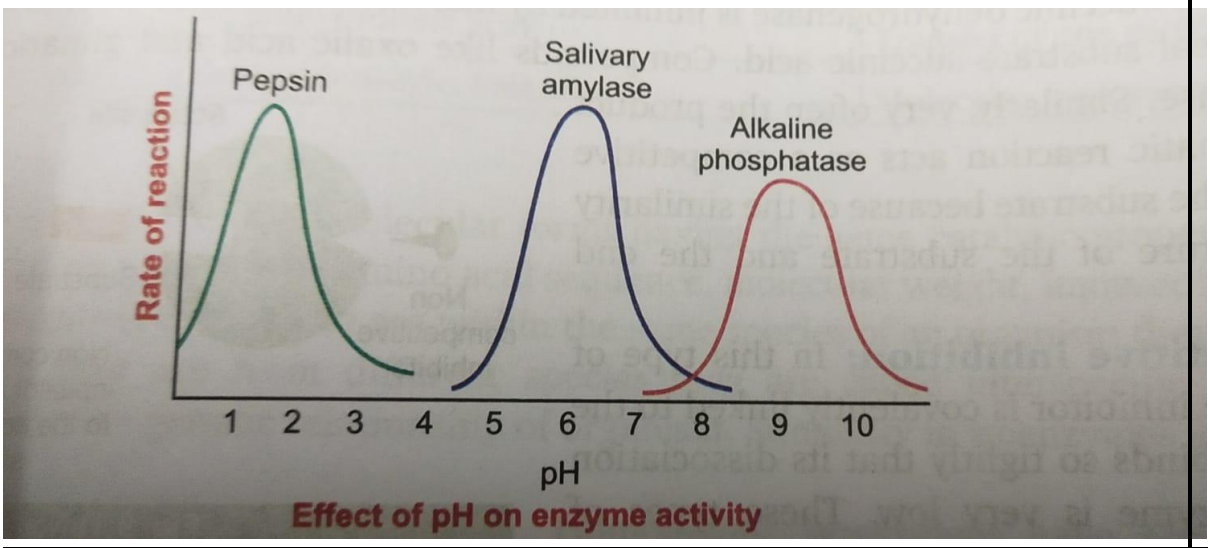


between a range of zero degree Celsius in

45 degrees Celsius the activity of enzymes increase with an increase in temperature. For every rise of 10°C, the rate of reaction is increased by two times, i.e., temperature coefficient is '2'. The optimum temperature for most of the enzymes is between 25°C and 45°C. At lower temperatures, the enzyme becomes inactivated. At higher temperatures the enzymes are destroyed and on lowering the temperature to optimum, the catalytic activity is not restored.

However, certain enzymes are very resistant to high temperature, for example, crystalline trypsin and taka-diastase are denatured reversibly even at 100°C.

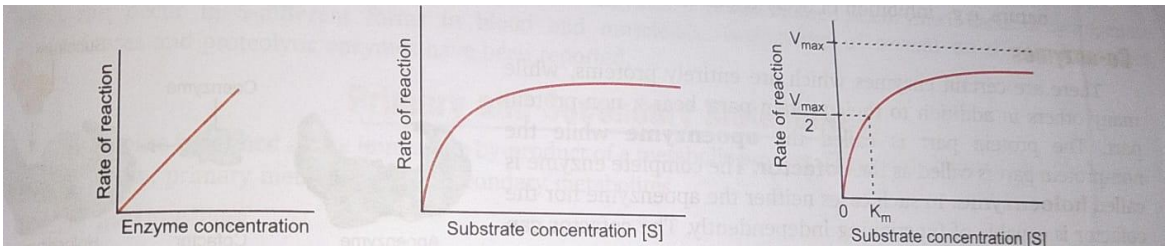
2.pH-



Effect of pH on enzyme activity

Each enzyme is catalytically active within a limited range of hydrogen ion concentration (pH). In other words each enzyme has its own specific pH optimum. For example, the optimum pH for pepsin is between 1.5 and 2.5 whereas for trypsin it is between 8 and 11. The activity of enzymes declines gradually on either side of the optimum.

3. Enzyme concentration-



The rate of a chemical reaction increases proportionately with increasing concentration of the enzyme provided that

substrate is present in an excess and the end products are continuously removed from the site of reaction.

4. <u>Substrate concentration-</u>

Substrate is the substance which is acted upon by the enzyme to produce the product. The catalytic activity of an enzyme is expressed in terms of its turnover number which is defined as the number of substrate molecules converted into product per unit time by an enzyme provided the substrate is in excess. Turnover numbers for most enzymes vary between 1 and 10^4 per second.

In the absence of substrate the enzyme will not show any activity. Now if the substrate concentration is increased gradually, the activity of enzyme increases and at a sufficiently high substrate concentration the maximum velocity (Vmax) of the reaction is achieved. If substrate concentration is increased beyond this point the rate of

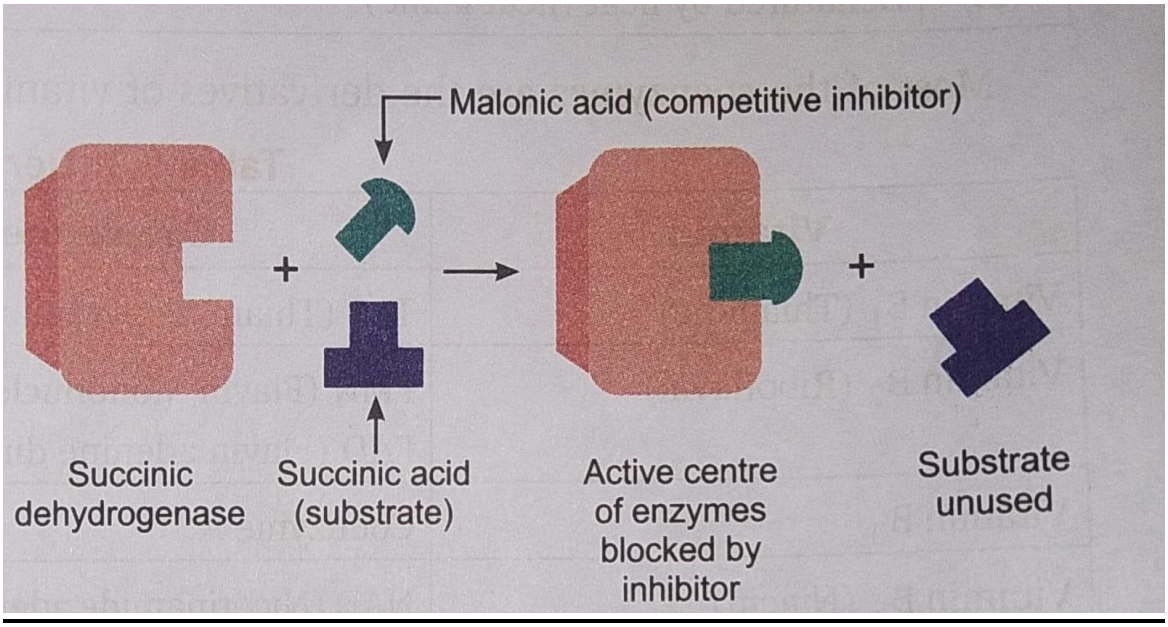
reaction does not show any appreciable increase; rather it may decline. At Vmax, all the enzyme molecules exist in the form of ES complex. The substrate concentration that produces half maximum velocity is termed Km value or Michaelis constant. It is expressed as:
|K|m= |V|max\2 {| | shows index}

5. **Concentration of products**- If the products of enzyme-catalysed reaction are allowed to accumulate in the reaction medium, the rate of reaction is retarded.

- **Enzyme Inhibition**

  The activity of enzymes can be inhibited by combination of substances with the enzymes which do not serve as substrates and are called as inhibitors. Inhibition is of following types:

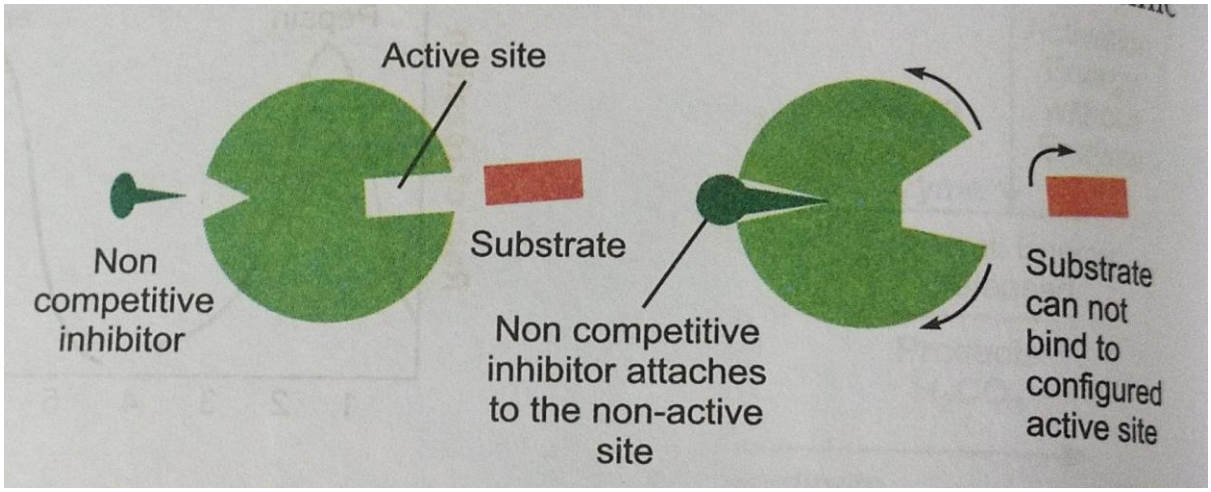
  1. Competitive or surface analog inhibition



Sometimes substances which are structurally similar (analogous) to the substrate molecules bind with the active site of an enzyme thus inhibiting the enzyme action. In other words these substances compete with the normal substrates for active sites on the enzyme. This type of inhibition is reversible and can be removed by increasing the concentration of the real substrate. Succinic dehydrogenase is inhibited by malonic acid which has a structure very similar to that of its real substrate succinic acid. Compounds like oxalic acid and glutaric acid also inhibit dehydrogenase. Similarly, the end product of an enzymatic reaction acts as a competitive

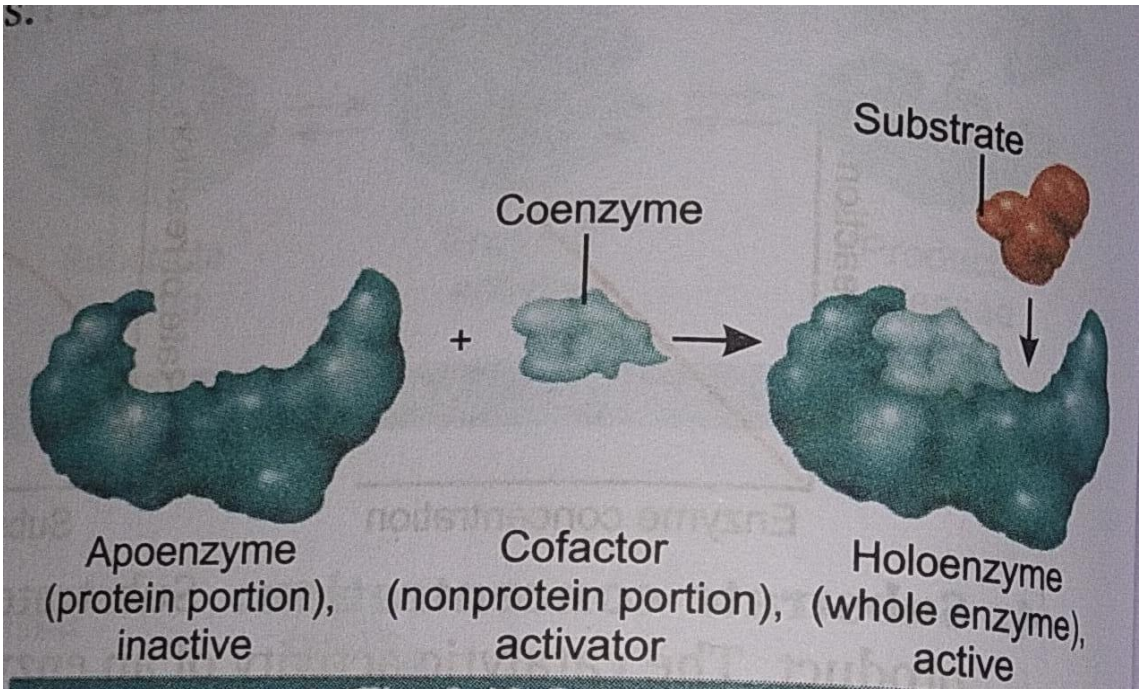
inhibitor because of the similarity in structure between the substrate and the end product.

2.Non -Competitive inhibition



In this type of inhibition the inhibitor is covalently linked to the enzyme and binds so tightly that its dissociation from the enzyme is very low. These types of inhibitors are either metabolic poisons or heavy metal ions which do not compete for the active site destroy the structure of enzymes this is irreversible in nature example inhibition of acetylcholinesterase by nerve gas.

CO-ENZYMES:



There are certain enzymes which are entirely proteins, while many others in addition to their protein part, bear a non- protein part. The protein part is called ‘apoenzyme’ while the non-protein part is called as the ‘cofactor’. The complete enzyme is called ‘holoenzyme’. In such cases neither the apoenzyme nor the cofactor is capable of functioning independently. The cofactor can be a prosthetic group or a co-enzyme. The prosthetic group remains firmly attached to the enzyme while taking part in the reaction.

For example, the porphyrin groups of cytochromes and biotin of carboxylase are tightly bound to their apoenzymes. When a metal functions as a prosthetic group it is generally referred to as an 'activator'. The coenzyme on the other hand remains loosely attached to the apoenzyme. These act as acceptor or donor often atom or a group that has been removed from or is to be added to a substrate ,that is why the Co - enzymes are also called as second substrates.

for example NAD,FAD,FMN and CoQ are hydrogen carriers, ATP donates phosphate group and CoA donates the acyl group.

DIFFERENCE BETWEEN APOENZYME AND CO-ENZYME :

APOENZYME	CO-ENZYME
MADE UP OF PROTIENS	USUALLY THESE ARE DERIVATIVES OF VITAMINS
HIGH MOLECULAR WEIGHT	LOW MOLECULAR WEIGHT

DENATURED BY HEAT	NOT AFFECTED BY HEAT

VITAMIN B- AN EXAMPLE OF CO-ENZYMES

VITAMIN	COENZYME FORM	TYPES OF REACTION CATALYZED
B1(THIAMINE)	TPP(THIAMINE PYROPHOSPHATE)	CARBOXYLATION REACTION
B2(RIBOFLAVIN)	FMN(FLAVIN MONONUCLEOTIDE) FAD(FLAVIN ADENINE DINUCLEOTIDE)	REDOX (OXIDATION REDUCTION) REACTION
B3	COENZYME A	ACETYL TRANSFER REACTION
B5(NIACIN)	NAD(NICOTINAMIDE ADENINE DINUCLEOTIDE)	DEHYDROGENASE REACTION
B6(PYRIDOXALAMINE)	PYRIDOXAL PHOSPHATE	TRANSAMINASE REACTION

BIOTIN	BIOCYTIN	CARBOXYLASE REACTION
FOLIC ACID	TETRAHYDROFOLIC ACID	TRANSFER OF ONE CARBON
B12	DEOXYADENOSYLCOBALAMINE	MOLECULAR REARRANGEMENT REACTION

Isoenzymes:

The enzymes which exist in multiple molecular forms having the same catalytic property are called isozymes or isoenzymes. They show differences in their amino acid sequence, molecular weight, immunological and electrophoretic behaviours. When the variants of an enzyme are within the same species of organism they are called intraspecific or ontogenic variants; when they are from different species they are called interspecific or phylogenetic variants. Isoenzymes provide a clue to the genetic relationship of organism. Similarity in isoenzymes is correlated with similarity in genotype.

Isoenzymes may be homologous or analogous. Homologous isoenzymes have essentially similar molecular structure and catalytic properties. Analogous isoenzymes, though catalyse similar reaction, have

different molecular structure and catalytic properties.

There are more than 100 enzymes which exist as isoenzymes. The best studied isoenzyme is lactic dehydrogenase which can occur in 5 different forms in blood and muscle-viz. Isoenzymes of several other dehydrogenases, transaminases and proteolytic enzymes have been reported.

NEWS ARTICLES ABOUT ENZYME ACTIVITY:

Scientists create enzyme that recycles plastic bottles in hours

Modified enzyme breaks down plastic into its basic building blocks

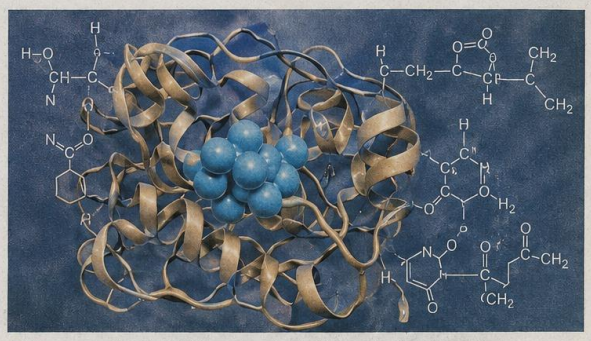
Damian Carrington Environment editor
Wed 8 Apr 2020 11.01 BST

Scientists have created a mutant enzyme that can break down plastic bottles in a matter of hours, which they say could help tackle the mountains of plastic polluting the planet.

The enzyme – a biological catalyst – eats into the plastic, PET, which is used to make bottles for water and soft drinks, breaking it down into its basic chemical building blocks.

PET is one of the biggest contributors to the plastic waste problem. It can take hundreds of years to break down in landfill or the ocean.

The researchers, from the University of Portsmouth and US-based biotech firm Viridos, said their enzyme could recycle PET waste at room temperature in just hours, rather than the days or weeks taken by existing enzymes.



▲ The enzyme breaks PET down into its basic chemical building blocks, which can then be reused. PHOTOGRAPH: UNIVERSITY OF PORTSMOUTH

The breakthrough raises the possibility of developing new ways to tackle plastic waste and move towards a circular economy in which materials can be used over and over again.

Dr John McGeehan, the lead researcher from the University of Portsmouth, said: "Our enzyme is incredibly efficient at break- PET. We hears ɛ thare does very slowly agd evolved it to do iy thousands of times faster." The enzyme was created using directed evolution – a technique that mimics nâtuiral selection in a laboratory.

The team tested millions of mutations of the enzyme to find versions that were better at breaking down PET. The best of these was then further refined to create a highly efficient enzyme.

HOW THE ENZYME WORKS

- The enzyme binds to PET plastic.
- It breaks the long polymer chains into smaller pieces.
- These pieces are then broken down into basic building blocks.
- The building blocks can be reused to make new plastic products.

"This is a significant step forward in addressing plastic pollution," said McGeehan. "While more work is needed before it can be used industrially, it shows great promise for a sustainable future."

The researchers are now working on scaling up the process and improving the stability of the enzyme for industrial use.

THE HINDU 9 APRIL 2020

Scientists create enzyme that recycles plastic bottles in hours

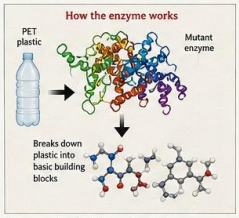
Breakthrough enzyme could help solve the mounting problem of plastic pollution

Agence France-Presse
London, April 8

Scientists have created a mutant enzyme that can break down plastic bottles in a matter of hours, a discovery that could help tackle the mountains of plastic polluting the planet.

The enzyme — a biological catalyst — eats into the plastic, PET, which is used to make bottles for water and soft drinks, breaking it down into its basic chemical building blocks.

The team, from the University of Portsmouth and US-based biotech firm Viridos, said their enzyme could recycle PET waste at room temperature in just hours, rather than the days or weeks taken by existing enzymes.



The enzyme breaks PET down into its basic chemical building blocks, which can be reused.
• UNIVERSITY OF PORTSMOUTH

THE TIMES OF INDIA 12 MAY 2023

Indian scientists develop artificial enzyme to combat dangerous blood clots

New 'nanozyme' shows promise in preventing thrombosis

TIMES NEWS NETWORK

Bengaluru: Scientists at the Indian Institute of Science (IISc), Bengaluru, have developed an artificial enzyme, or 'nanozyme', that can effectively prevent the formation of blood clots, which are a major cause of heart attacks and strokes.

The nanozyme mimics the natural enzyme found in our bodies and helps in the breakdown of harmful molecules that trigger clot formation.

The researchers say the discovery could lead to the development of safer and more effective anti-clotting drugs.



The nanozyme helps in dissolving clots and may revolutionise treatment for cardiovascular diseases.

The Indian EXPRESS 4 AUGUST 2022

New enzyme converts agricultural waste into clean biofuel

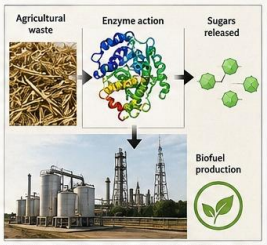
Discovery may boost sustainable energy production

EXPRESS NEWS SERVICE
New Delhi, August 3

A team of Brazilian scientists has discovered a new enzyme that can break down plant waste more efficiently, converting it into sugar that can be used to produce biofuel.

The enzyme, named CelOCE, worked 50% faster than other similar enzymes in laboratory tests.

The discovery can help reduce the cost of biofuel production and contribute to a cleaner and greener environment.



The enzyme breaks down complex plant materials into simple sugars, which are used to produce biofuel.

Hindustan Times 21 SEPTEMBER 2023

AI tool predicts enzyme functions with high accuracy

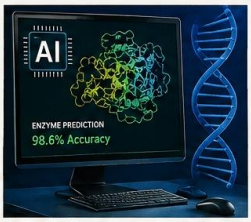
Breakthrough can accelerate drug discovery and biotechnology

HT Correspondent
New Delhi

Scientists have developed an artificial intelligence (AI) tool that can predict the functions of enzymes with high accuracy just by analysing their amino acid sequences.

The tool called CLEAN was developed by researchers at tUniversity of Illinois Urbana-Champaign.

This advancement can significantly speed up research in medicine, agriculture, and industrial biotechnology.



The AI model analyses protein sequences and predicts how enzymes will function. • UNIVERSITY OF ILLINOIS

THE ECONOMIC TIMES 15 JANUARY 2021

Industrial enzymes improving food quality and reducing costs

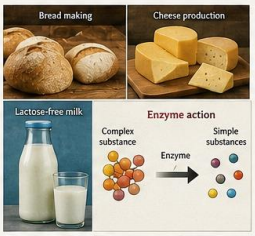
From bread to cheese, enzymes play a key role in food industry

ET Bureau
Mumbai, Jan 14

Enzymes are increasingly being used in the food industry to improve quality, enhance flavour and reduce production costs.

They are used in the production of bread, cheese, fruit juice, wine and even in lactose-free milk.

Enzymes help in breaking down complex substances into simpler forms, making food easier to digest and increasing shelf life.



Enzymes are biological catalysts that make food processing more efficient and consumer-friendly.

DNA 6 MARCH 2022

New thermostable enzyme works efficiently even at high temperatures

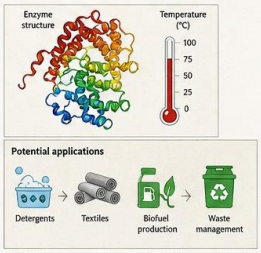
Discovery opens doors for industrial and environmental applications

DNA Correspondent
Pune

Researchers have discovered a new heat-resistant enzyme that remains stable and active even at high temperatures.

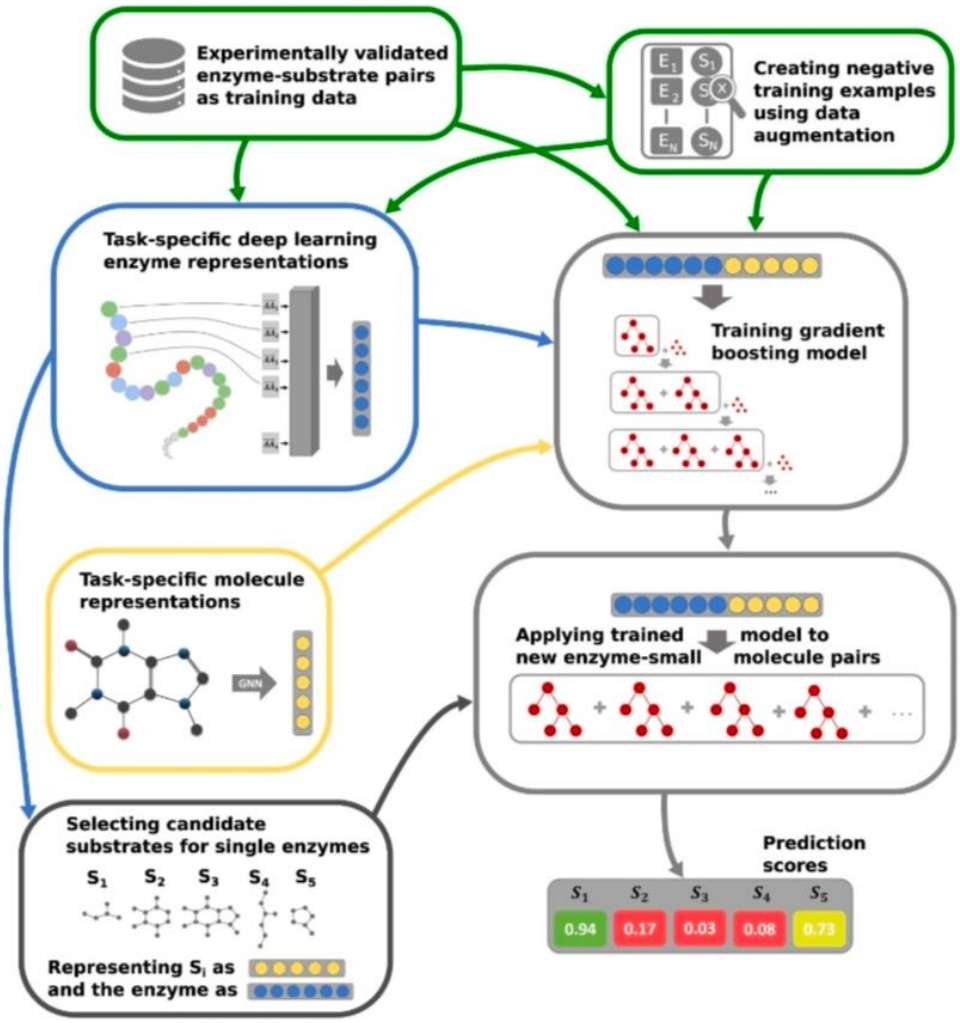
The enzyme can be used in industries such as detergents, textiles, biofuel production and waste management.

Its ability to function in extreme conditions can improve efficiency and reduce energy costs.



The enzyme retains its activity at high temperatures, making it ideal for various industrial processes.

AI AND ENZYME ACTIVITY :



This AI workflow predicts enzyme-substrate specificity using machine learning. First, the

system collects validated enzyme-substrate pairs alongside augmented negative training data. Deep learning models then convert amino acid sequences into mathematical enzyme representations. Concurrently, Graph Neural Networks (GNNs) generate molecular representations for the target substrates. A gradient boosting model trains on these combined feature vectors to learn binding patterns. Finally, the trained model evaluates novel enzyme-small molecule pairs. It outputs quantitative prediction scores to identify the most viable candidate substrates.

FINAL CONCLUSION:

The experiments demonstrate that enzyme activity depends on environmental conditions. Catalase exhibits maximum activity at an optimum temperature and neutral pH. Any deviation from these optimum conditions decreases enzyme activity because the structure of the enzyme and its active site are affected. Thus, temperature and pH are important factors controlling enzymatic reactions in living organisms.

FUTURE SCOPE:

The present study investigated the effects of temperature and pH on the activity of catalase in potato tissue. Future studies may expand upon this research by examining additional factors that influence enzyme activity, such as substrate concentration, enzyme concentration, and the presence of inhibitors.

Further investigations could involve quantitative measurement of oxygen production using more advanced laboratory equipment to obtain precise results. Similar studies may also be conducted using catalase from different biological sources, such as liver tissue, yeast, or plant leaves, to compare enzyme activity across organisms.

Understanding enzyme behavior under varying conditions has important applications in biochemistry, biotechnology, medicine, and industrial processes. Future research in this field

may contribute to a deeper understanding of enzymatic reactions and their practical applications.

COMPUTATIONAL ANALYSIS OF CATALASE ACTIVITY USING PYTHON

- **RESEARCH QUESTION :**

HOW CAN PYTHON-BASED COMPUTATIONAL ANALYSIS BE USED TO PROCESS AND VISUALIZE THE EFFECT OF TEMPERATURE AND PH ON POTATO CATALASE ACTIVITY?

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1. TEMPERATURE EXPERIMENT :

COMPUTATIONAL QUESTION:

How does temperature affect the measured rate of oxygen production by potato catalase?

- Python can be used to:

Organise the temperature and experimental measurements;

Calculate the reaction rate from recorded oxygen production and time;

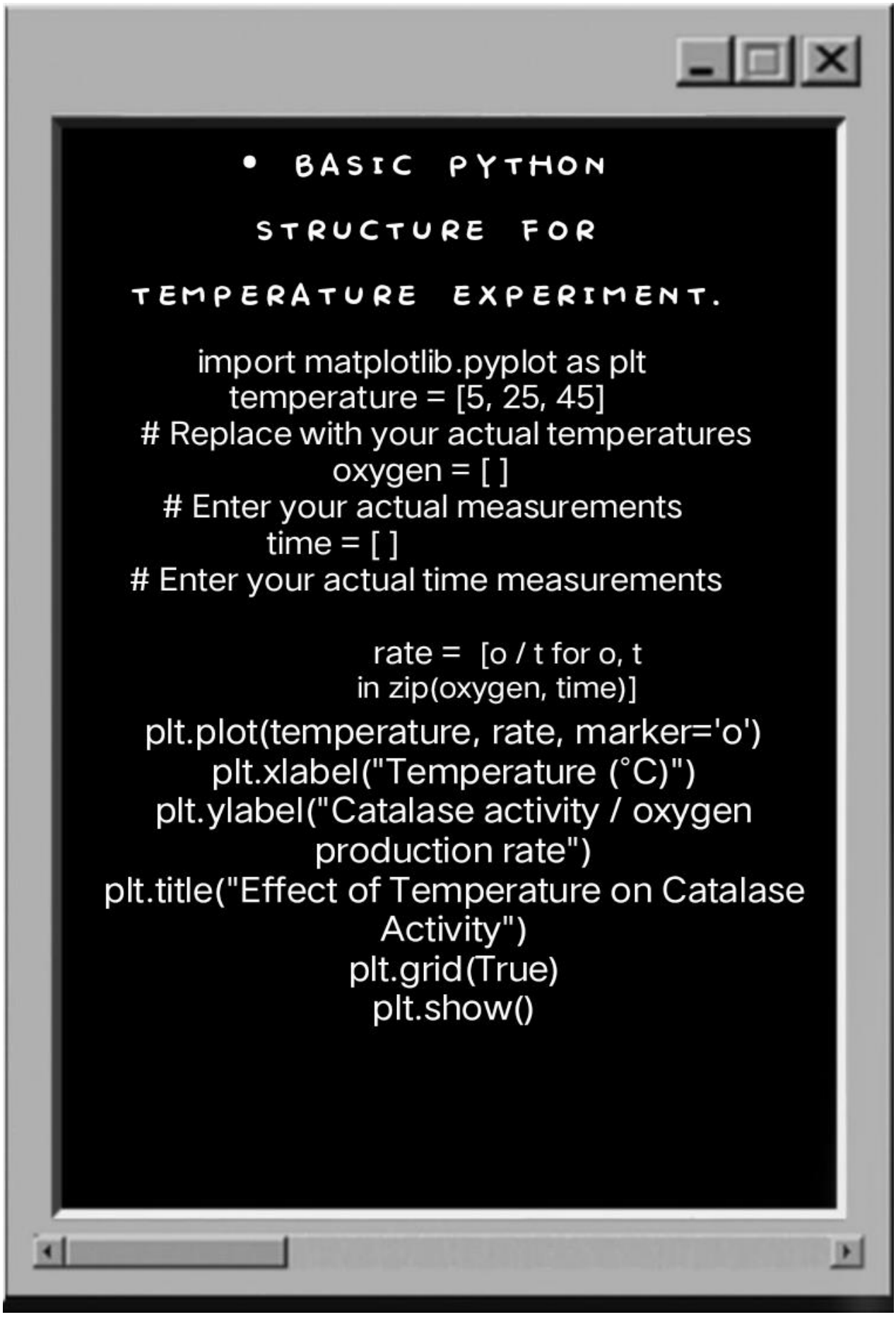
Compare catalyst activity at different temperatures;

Plot temperature vs catalase activity;

Identify the temperature at which the highest measured activity occurred.

- Here’s the basic structure written in Python for temperature experiment:

  This Python program uses the matplotlib library to analyse and visualise experimental data regarding the effect of temperature on catalase enzyme activity . it calculates the rate of oxygen production across different temperatures and generates a clear line graph to present the results.


• BASIC PYTHON
STRUCTURE FOR
TEMPERATURE EXPERIMENT.
import matplotlib.pyplot as plt
temperature = [5, 25, 45]
Replace with your actual temperatures
oxygen = [ ]
Enter your actual measurements
time = [ ]
Enter your actual time measurements
rate = [o / t for o, t
in zip(oxygen, time)]
plt.plot(temperature, rate, marker='o')
plt.xlabel("Temperature (°C)")
plt.ylabel("Catalase activity / oxygen
production rate")
plt.title("Effect of Temperature on Catalase
Activity")
plt.grid(True)
plt.show()

2. PH EXPERIMENT :

COMPUTATIONAL QUESTION:

How does pH affect the measure rate of oxygen by potato catalase?

- Python can similarly be used to:
  Organise the pH measurements;
  Calculate the reaction rates ;
  Compare activity at different pH values;
  Plot PH vs catalase activity;
  identify the pH associated with the highest measured activity.

- Here's the basic structure written in Python for PH experiment:

This Python script is designed to calculate analyse and visualise experimental data investigating how varying Ph levels affect the catalytic activity of enzyme catalase. The same function from the library that is matplotlib is used as used in temperature experiment.

• BASIC PYTHON STRUCTURE FOR pH EXPERIMENT

```
import matplotlib.pyplot as plt

pH = [4, 7, 9]
# Replace with your actual pH values
oxygen = [ ]
# Enter your actual measurements
time = [ ]
# Enter your actual time measurements

rate =  [o / t for o,
t in zip(oxygen, time)]

plt.plot(pH, rate, marker='o')
plt.xlabel("pH")
plt.ylabel("Catalase activity / oxygen
production rate")
plt.title("Effect of pH on Catalase Activity")
plt.grid(True)
plt.show()
```

3. INTERPRETATION:

The computation analysis provides a quantitative extension of this experimental investigation. Python was used to organise the experimental observations calculate activity where numerical measurements were available and, visualise the relationship between environmental conditions and catalase activity. Temperature and Ph analyses can then be compared with the experimental observations to determine how changes in environmental conditions affect catalase activity.

ACKNOWLEDGEMENT:

I would like to express my sincere gratitude to everyone who supported me during this research project. I am thankful to my teachers for their guidance and encouragement, and to my family for their support throughout the study. Their assistance and motivation contributed significantly to the successful completion of this project.

REFERENCES:

- BIOLOGY TEXTBOOKS
- MY BIOLOGY TEACHER
- BRITANICA ARTICLES ON ENZYMES
- LEHNINGER PRINCIPLES OF BIOCHEMISTRY


Thank you!